

In-Country Working Session: FASTR Implementation & RMNCAH-N Service Monitoring Analysis

January 27-30, 2026 | Lusaka

GFF FASTR Team

Agenda

Day 1 -- Laying the Foundation: Introducing FASTR and Configuring the Analytics Platform

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30-09:00	Participant registration	MoH team
09:00-09:10	Welcome and opening remarks	MoH team
09:10-09:20	Icebreakers/Introductions	MoH team
09:20-09:35	Overview of agenda, workshop objectives	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35-10:30	Overview: FASTR Approaches	GFF FASTR team
Session 2: HMIS data extraction		
10:30-11:30	Data extraction: Rationale and methods	GFF FASTR team
Session 3: Introduction to the FASTR analytics platform		
11:30-12:30	Introduction to the FASTR analytics platform	GFF FASTR team
12:30-14:00	<i>Lunch Break</i>	
14:00-14:30	Getting participants into the platform	GFF FASTR team
Session 4: Configuring the FASTR analytics platform		
14:30-16:30	Configuring the analysis platform	GFF FASTR team

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Day 2 -- Building the Analysis: Applying FASTR Methods and Generating Outputs

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 2 agenda	GFF FASTR team
Session 5: Overview of FASTR methods and analytical outputs		
09:15-10:15	Data quality, service utilization, coverage	GFF FASTR team
Session 6: Creating a project		
10:15-11:15	Project creation and settings	GFF FASTR team
Session 7: Creating visualizations		
11:15-12:30	Creating and editing visualizations	GFF FASTR team
12:30-14:00	<i>Lunch Break</i>	
Session 8: Creating reports		
14:30-16:30	Practice creating and editing reports	GFF FASTR team

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Day 3 -- From Analysis to Action: Interpreting Results and Using FASTR for Decision-Making

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 3 agenda	GFF FASTR team
Session 8: Interpretation of visualizations		
09:15-10:15	Approaches to support interpretation	GFF FASTR team
Session 9: Creating a Q4 2025 report		
10:15-12:30	Creating short and long reports with country context	GFF FASTR team
12:30-14:00	<i>Lunch Break</i>	
Session 9 (cont'd): Creating a Q4 2025 report		
14:00-15:00	Continue report creation with country context	GFF FASTR team
Session 10: Presenting reports		
14:30-15:30	Present reports, group feedback	GFF FASTR team

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Day 4 -- Designing the Health Facility Assessment

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 4 agenda	GFF FASTR team
Session 12: Overview of FASTR HFA phone survey		
09:15-10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 13: Questionnaire adaptation to the Zambian context		
10:15-12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30-14:00	<i>Lunch Break</i>	
Session 14: Questionnaire adaptation (cont'd)		
14:00-15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 15: Discussion: HFA adapted questionnaire		
14:30-15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 16: Discussion: HFA priorities and data use		
15:30-16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 17: Action planning and wrap-up		
16:30-17:00	Key messages and wrap-up	GFF FASTR team

Workshop Objectives

- Prepare MoH for participation in the multi-country workshop
- Strengthen capacity for disruption analysis and data extraction
- Configure the FASTR analytics platform for Zambia
- Produce the first quarterly report
- Adapt the phone survey questionnaire to the Zambian context
- Use data downloader tools for DHIS2

Scope of Work

| Disruption Analysis Activities

- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)

| Phone Survey Activities

- Adaptation of questionnaire (with program input)

| Capacity Building

- Training on data downloader for all with DHIS2 access
- Hybrid format: face-to-face and online

Expected Outputs

- Adapted phone survey questionnaire
- First quarterly report draft
- Trained MoH team on data downloader tools
- Roadmap for participation in Abuja workshop

Session 1: Overview of the FASTR approach

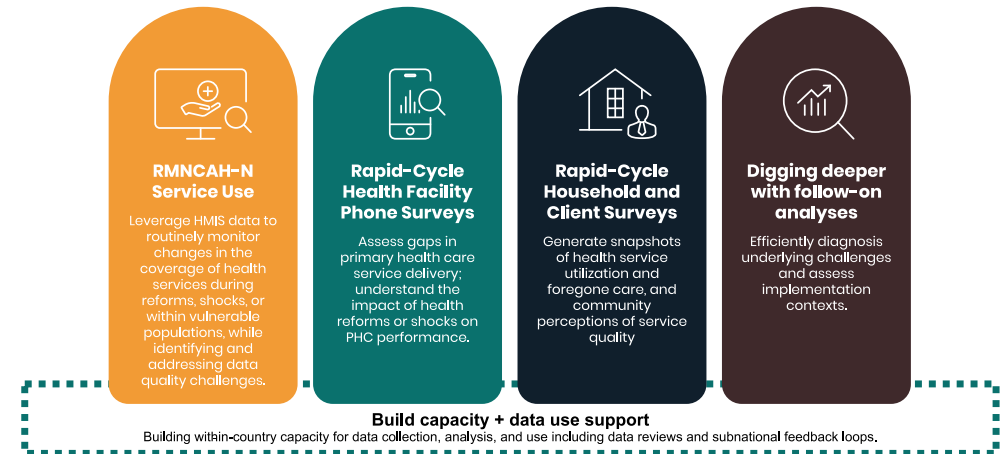
Introduction to FASTR

The Global Financing Facility (GFF) supports country-led efforts to strengthen the use of timely data for decision-making, with the goal of improving primary healthcare (PHC) performance and RMNCAH-N outcomes.

Frequent Assessments and Health System Tools for Resilience (FASTR) is the GFF's rapid-cycle analytics framework for monitoring health system performance using high-frequency data.

FASTR brings together four complementary technical approaches:

1. Routine HMIS data analysis
2. Health facility phone surveys
3. High-frequency household phone surveys
4. Follow-on, problem-driven analyses



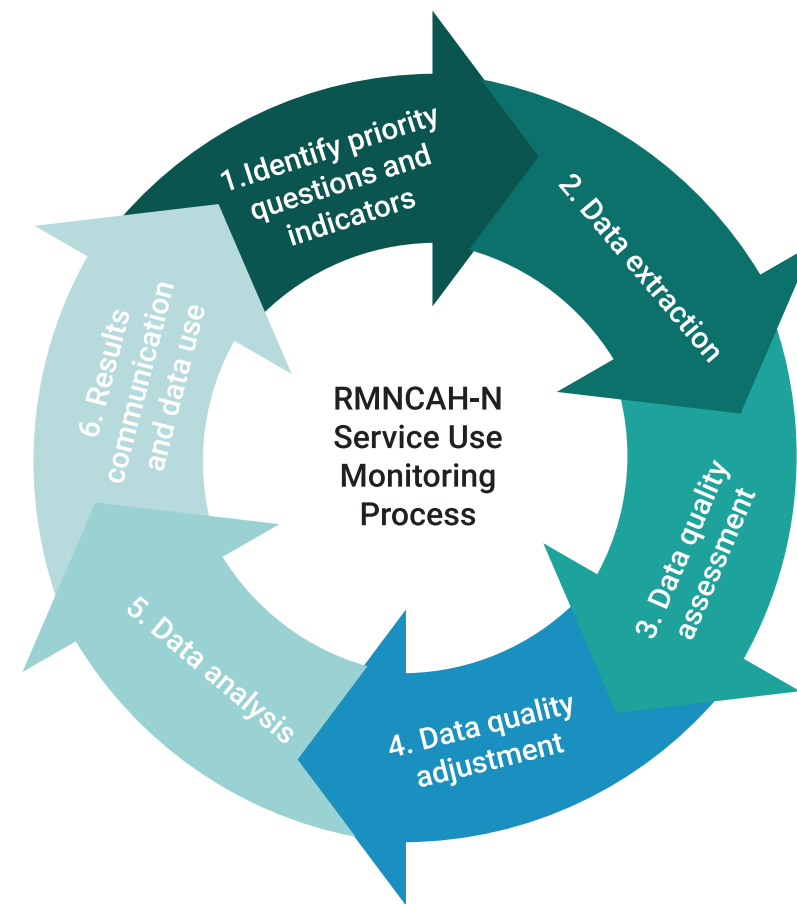
RMNCAH-N service use monitoring

Rapid-cycle approaches using routine health management information system (HMIS) data can be used to:

- **Evaluate the quality of HMIS data** both at the national and sub-national levels to help address gaps in quality and completeness
- **Measure monthly changes** in the utilization of critical health services, enabling a rapid response to challenges and the opportunity to gain insights on the progress of reforms
- **Compare service coverage trends** with country targets, facilitating the continuous monitoring of RMNCAH-N progress

| How does it work?

Through collaboration with Ministries of Health, the GFF assists countries in developing and reviewing regular analyses of HMIS data, focusing on specific priority health service indicators linked to national health care reforms, as well as GFF and World Bank investments.



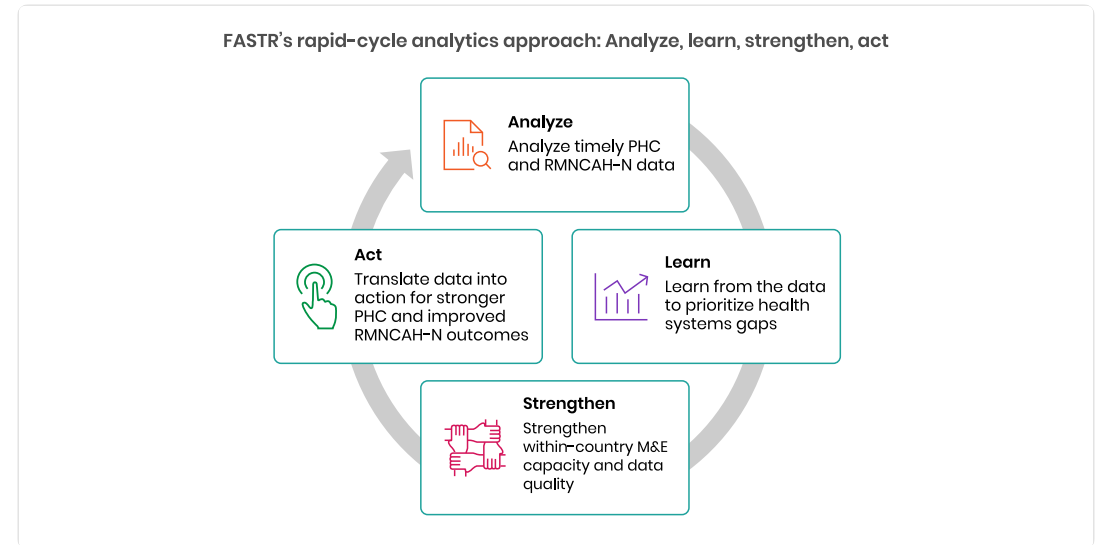
Why rapid-cycle analytics?

Routine health information systems are a critical source of data, but they are often underused due to concerns about data quality and long delays between data collection and analysis. Traditional household and facility surveys, while essential, are resource-intensive and infrequent.

FASTR's rapid-cycle analytics address this gap by providing:

- Timely insights aligned with country decision cycles
- Continuous learning rather than one-off assessments
- Direct feedback loops between data, analysis, and action

During the COVID-19 pandemic, this approach was applied in over 20 countries to monitor disruptions to essential RMNCAH-N services and inform response and recovery planning.



Focus of the analysis

| Core indicators

FASTR prioritizes a core set of RMNCAH-N indicators that:

- Represent key service delivery contacts across the continuum of care
- Have relatively high reporting completeness and volumes
- Serve as proxies for broader service delivery performance

Outpatient consultations are included as a proxy for overall health service use. The indicator set can be expanded to reflect country-specific priorities.

| Core data quality metrics

Analysis is anchored in a standardized set of data quality metrics, including:

- Reporting completeness
- Extreme value (outlier) detection
- Consistency across related indicators

These metrics are summarized into an overall data quality score to support interpretation and comparison across areas.

Session 2: HMIS data extraction

Why extract data from DHIS2?

| Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

The FASTR methodology includes specific approaches to:

- Identify and adjust for outliers
- Adjust for incomplete reporting
- Apply consistent data quality metrics

These adjustments require processing that cannot be done within DHIS2's native analytics.

Why extract data from DHIS2?

| Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by:

- Identifying significant increases or decreases in service volume
- Adjusting for data quality issues
- Accounting for expected seasonal variations
- Comparing key periods, such as before and after a reform

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis.

Data format and granularity

Data should be downloaded for each **indicator of interest**, at **facility level**, and **monthly** for the **period of interest**.

- Data should be saved in **long format** meaning each row represents a single observation or measurement
- Data should be saved in **.csv format** and can be saved in either a single .csv file or multiple .csv files

| Why monthly facility level data?

We want to use the most granular data we have access to in order to make more fine tuned assessments for data quality. Using monthly facility level data allows us to conduct the most robust analysis.

Key variables

The data extracted should include the following required elements:

Element	Description
Org units	Organizational unit identifier
Period	Time period of the data
Indicator name	Name of the indicator
Total/count	The aggregated value

How much data?

| Initial FASTR analysis

- Download approximately **five years** of historical data
- Exact period depends on data availability and consistency in indicator definitions

| Routine update to FASTR analysis

- Download new data covering the most recent months not previously included (usually **three months** for quarterly implementation)
- Include the **three preceding months** as recent data is often subject to changes due to late reporting or data quality adjustments

Data extraction tools

We offer two tools for bulk DHIS2 data extraction:

API Script (Google Colab)

- Input login credentials, specify timeframes, indicators, and administrative levels
- Download data as a .csv file

Data Downloader

- More intuitive, streamlined interface
- Recommended for most users

Both tools enable efficient data extraction, and we provide training resources to support their use.

DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

Key features:

- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

Download from GitHub:

<https://github.com/worldbank/DHIS2-Downloader/releases/>

Facilitator will demonstrate the Data Downloader

Session 3: Introduction to the FASTR analytics platform

Introduction to the FASTR Analytics Platform

The FASTR analytics platform is a web-based tool for data quality assessment, adjustment, and analysis of routine health data.

Key features:

- Upload and analyze data from DHIS2 and other sources
- Built-in statistical methods for data quality adjustment
- User-friendly interface for running analyses
- Flexible visualization and export options

In this session, we will provide a conceptual walkthrough of the platform and its capabilities.

Live Demo: Platform Access & Roles

In this demo, we will:

- Navigate to the FASTR platform
- Explore user roles: Administrator, Editor, Viewer
- Review user management and permissions
- Understand the workflow for uploading data and making analytical decisions

Facilitator will demonstrate in the live platform



Lunch Break

90 minutes

Back at 14:00

Session 4: Configuring the FASTR analytics platform

Activity: Setting Up Admin Areas

In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data

In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules

In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Day 2

Recap

On Day 1, we covered:

- The FASTR approach and rapid-cycle analytics methodology
- Data extraction from DHIS2 using the Data Downloader
- Introduction to the FASTR Analytics Platform
- Platform configuration: admin areas, data import, modules

Day 2 Focus

Today we will:

- Explore FASTR methods for data quality and analysis
- Create and configure projects
- Build visualizations
- Generate reports

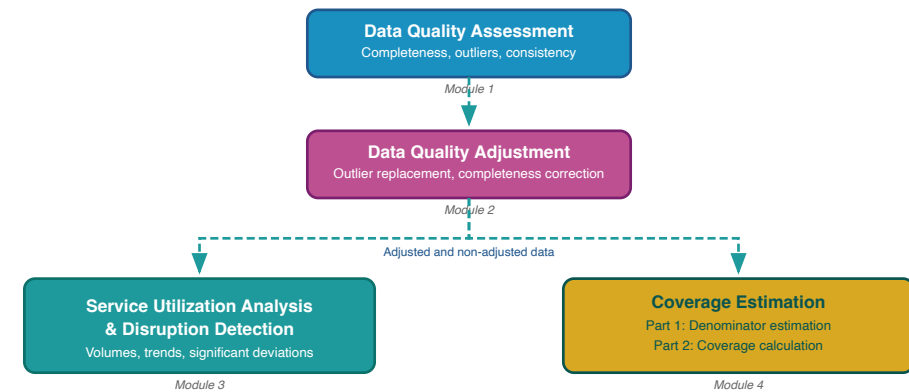
Session 5: Overview of FASTR methods and analytical outputs

FASTR analytical pipeline

The FASTR analysis follows a sequential workflow where each step builds on the previous:

- 1. Assess data quality** - Identify issues with completeness, outliers, and consistency
- 2. Adjust for quality issues** - Apply corrections to improve data reliability
- 3. Analyze adjusted data** - Generate service utilization and coverage estimates

Each step must be completed before moving to the next.



Data quality assessment

Evaluating the reliability of routine health information system data

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Objectives of data quality assessment

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Core dimensions of data quality

1. Completeness

Are health facilities submitting reports consistently?

2. Outlier prevalence

Are reported values within plausible ranges?

3. Internal consistency

Do related indicators demonstrate expected relationships?

These three dimensions provide a comprehensive assessment of data reliability for analytical purposes.

Assessing reporting completeness

Definition:

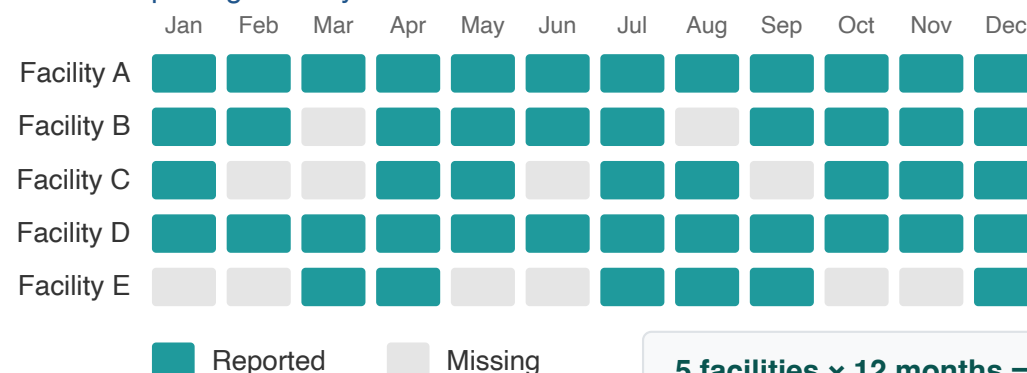
The proportion of expected facility reports that were submitted within a given period.

Significance:

- Incomplete reporting limits the representativeness of aggregate data
- Apparent declines in service volumes may reflect reporting gaps rather than actual reductions

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities × 12 months = 60 expected
48 reported → 80% completeness

Interpreting completeness rates

Reference benchmarks:

- 90% and above: High completeness
- 80-89%: Moderate completeness
- Below 80%: Substantial reporting gaps

Considerations: Complete reporting does not capture services delivered outside the formal health system or by facilities not included in the HMIS.

Key analytical questions: Are completeness rates improving over time? Which geographic areas or facility types demonstrate lower completeness?

Completeness: FASTR output

Indicator Completeness

Percentage of facility - months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

	90% or above
	80% to 89%
	Below 80%

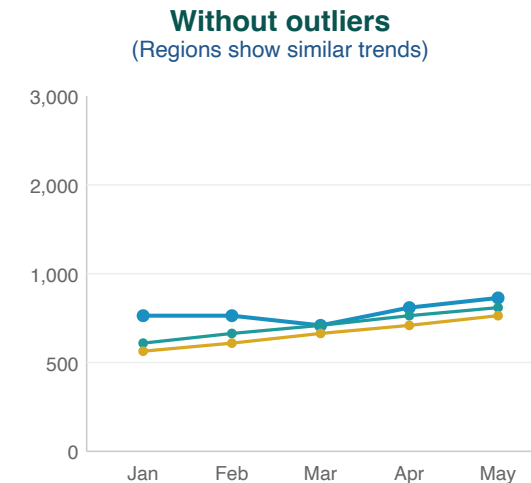
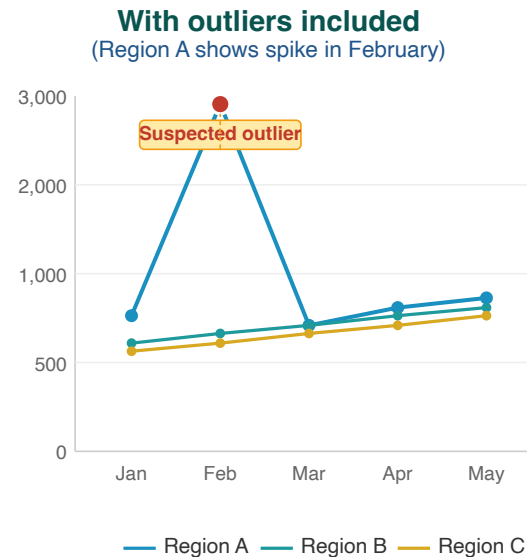
Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Identifying implausible values

Illustration:

Region A displays an anomalous increase in February that substantially exceeds values reported by other regions.

This pattern is indicative of a data entry error. Following adjustment, all regions demonstrate consistent gradual trends.



Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified through analysis of within-facility variation in monthly reporting for each indicator.

A value is classified as an outlier if it meets EITHER criterion:

1. The value exceeds 10 times the Median Absolute Deviation (MAD) from the facility's monthly median for that indicator, OR
2. The value represents more than 80% of the total volume for a given facility, indicator, and time period

AND the reported count exceeds 100.

Outlier illustration

Health Centre B - Malaria diagnostic tests:

Month	Tests reported	Classification
January	245	Within expected range
February	267	Within expected range
March	2,890	Outlier
April	256	Within expected range

Probable cause: Data entry error (e.g., "2890" entered instead of "289")

Analytical impact: Without adjustment, the data would indicate an erroneous increase in malaria testing during March.

Outlier prevalence: FASTR output

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

	3% or above
	1% to 2%
	Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Assessing relationships between indicators

Related health services demonstrate predictable relationships that can be used to assess data quality. FASTR evaluates the following indicator pairs:

Indicator pair	Expected relationship	Rationale
ANC1 / ANC4	$ANC1 \geq ANC4$	More women initiate antenatal care than complete four visits
Penta1 / Penta3	$Penta1 \geq Penta3$	More children receive the first dose than complete the series
BCG / Deliveries	$BCG \approx Deliveries$	BCG is administered at birth; counts should be similar

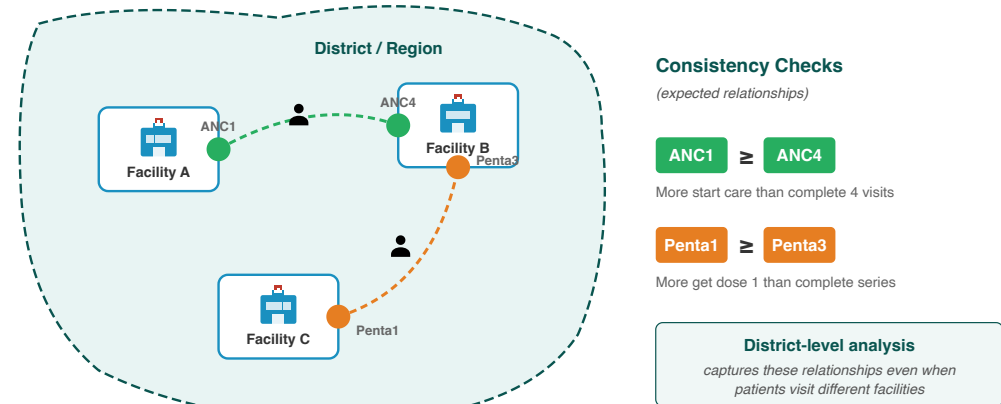
Violations of these expected relationships indicate potential data quality issues requiring investigation.

Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughly the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Calculating the overall quality score

The composite score integrates all three data quality dimensions:

1. **Completeness:** Did the facility submit a report?
2. **Outlier status:** Are reported values within plausible ranges?
3. **Consistency:** Do related indicators demonstrate expected relationships?

Binary DQA score:

- Score = 1 if all three criteria are satisfied
- Score = 0 if any criterion is not met

Mean DQA score: Weighted average of completeness-outlier score and consistency score

Applications:

- Inform decisions regarding data inclusion in analyses
- Identify facilities requiring targeted data quality support

Overall DQA score: FASTR output

Overall DQA score

Percentage of facility - months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

- 80% or above
- 70% to 79%
- Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: FASTR output

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

- 80% or above
- 70% to 79%
- Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

DQA module: Configuration parameters

The Data Quality Assessment module uses configurable parameters:

Parameter	Default	Description
GEOLEVEL	admin_area_3	Geographic level for consistency analysis
OUTLIER_PROPORTION_THRESHOLD	0.80	Flag if single month exceeds this proportion of annual volume
MINIMUM_COUNT_THRESHOLD	100	Only flag outliers with count above this threshold
MADS	10	Number of MADs for statistical outlier detection
DQA_INDICATORS	penta1, anc1, opd	Core indicators for DQA scoring
CONSISTENCY_PAIRS_USED	penta, anc	Indicator pairs for consistency assessment

Tuning DQA parameters

Objective	Adjustment
More sensitive outlier detection	Lower <code>MADS</code> to 8; lower <code>OUTLIER_PROPORTION_THRESHOLD</code> to 0.6–0.7
Less sensitive outlier detection	Increase <code>MADS</code> to 12–15; increase <code>OUTLIER_PROPORTION_THRESHOLD</code> to 0.9
Include small facilities	Lower <code>MINIMUM_COUNT_THRESHOLD</code> to 50
Focus on large facilities	Increase <code>MINIMUM_COUNT_THRESHOLD</code> to 200+
Coarser consistency analysis	Set <code>GEOLEVEL</code> to <code>admin_area_2</code>

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from outlier adjustment

Adjustment module: Configuration parameters

The Data Quality Adjustment module uses these key parameters:

Parameter	Default	Description
Rolling window	6 months	Window size for calculating replacement values
Low volume threshold	100	Indicators never exceeding this count are excluded from outlier adjustment

Adjustment hierarchy

When replacing outliers or filling missing values, the module applies methods in priority order:

Priority	Method	When used
1	Centered 6-month average	Default: 3 months before + 3 months after
2	Forward 6-month average	When insufficient preceding data
3	Backward 6-month average	When insufficient following data
4	Same month, previous year	For seasonal indicators when rolling averages unavailable
5	Facility historical mean	Fallback when no other method available

Excluded indicators: Mortality-related indicators (maternal deaths, neonatal deaths, under-5 deaths) are never adjusted.

Service utilization analysis

Analysis of health service delivery patterns to detect and quantify changes in service volumes over time.

Objectives

1. Measure changes in service volume

Year-over-year comparison of service volumes identifies increases or decreases across regions and indicators.

2. Detect and quantify disruptions

Statistical comparison of observed volumes against expected levels—derived from historical trends and seasonal patterns—enables identification and quantification of service shortfalls or surpluses.

Year-over-year change

Year-over-year percent change quantifies shifts in service delivery between consecutive years.

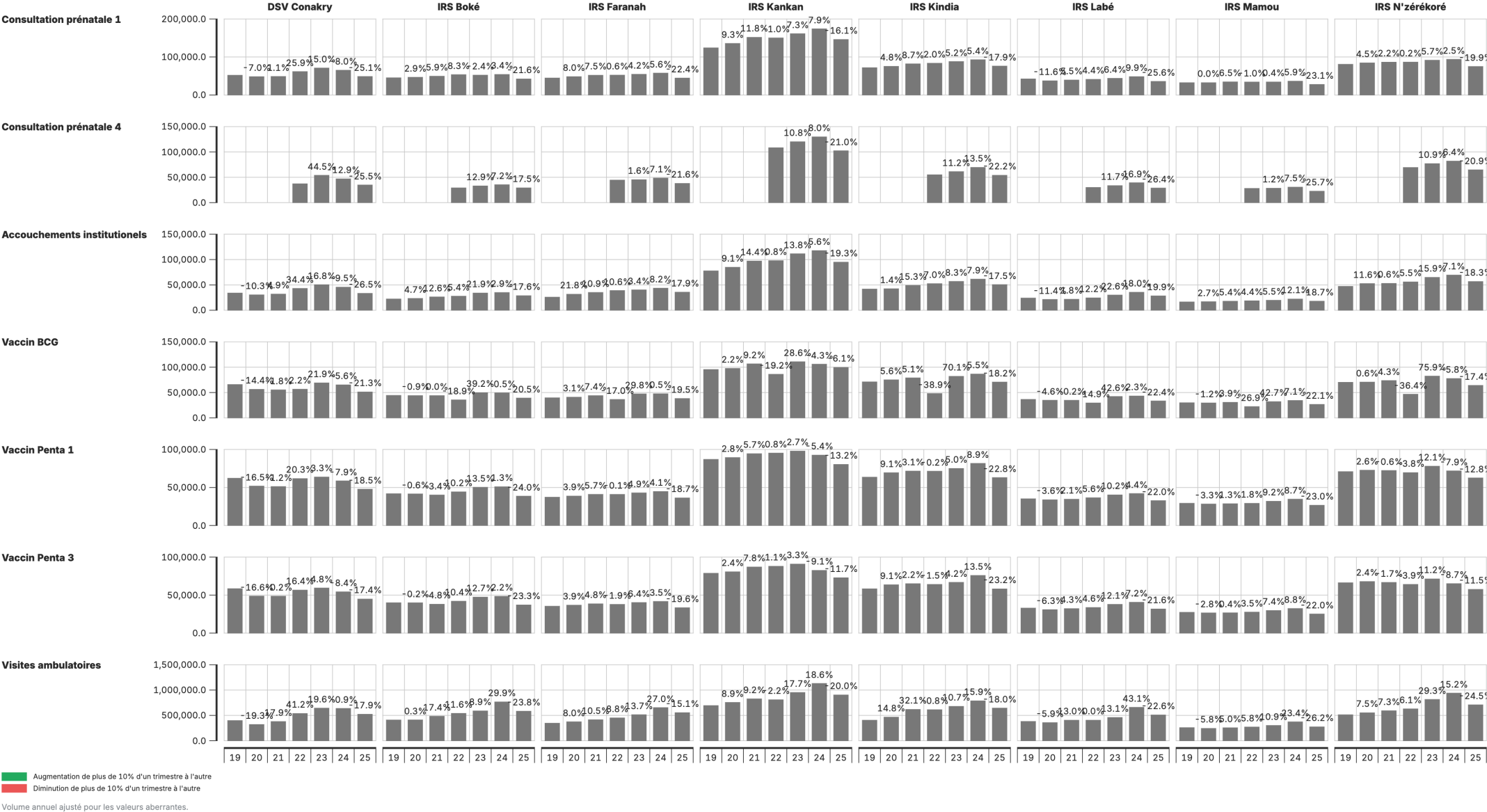
For each indicator and region, total volume in the current year is compared to the previous year:

$$\text{Percent change} = \frac{\text{Current year} - \text{Previous year}}{\text{Previous year}} \times 100$$

Changes exceeding **±10%** are flagged for review.

Service volume by year & year-on-year change

Janv 2019 à Sept 2025



Interpretation

Element	Description
Bars	Annual service volumes by region
Percentages	Year-over-year change annotations

Key considerations:

- Which regions exhibit the largest changes?
- Are changes consistent across regions or geographically concentrated?
- Do patterns vary by indicator?

Disruption detection

The analysis identifies periods where service delivery deviates significantly from expected levels using a two-stage approach:

Stage	Method	Purpose
1. Control chart analysis	Statistical process control	Identify when disruptions occur
2. Regression analysis	Panel regression models	Quantify disruption magnitude

Expected service volume

Expected volume is modeled using two components:

Component	Description
Trend	Long-term direction (increasing, decreasing, or stable)
Seasonality	Predictable monthly patterns (e.g., increased malaria cases during rainy season)

The model estimates expected volume for each month based on historical patterns. Significant deviations from expected values are flagged as potential disruptions.

Shortfalls and surpluses

Comparison	Interpretation
Actual = Expected	Services delivered as anticipated
Actual < Expected	Shortfall: fewer services than expected
Actual > Expected	Surplus: more services than expected

Surpluses may indicate catch-up campaigns, outreach activities, or reporting changes.

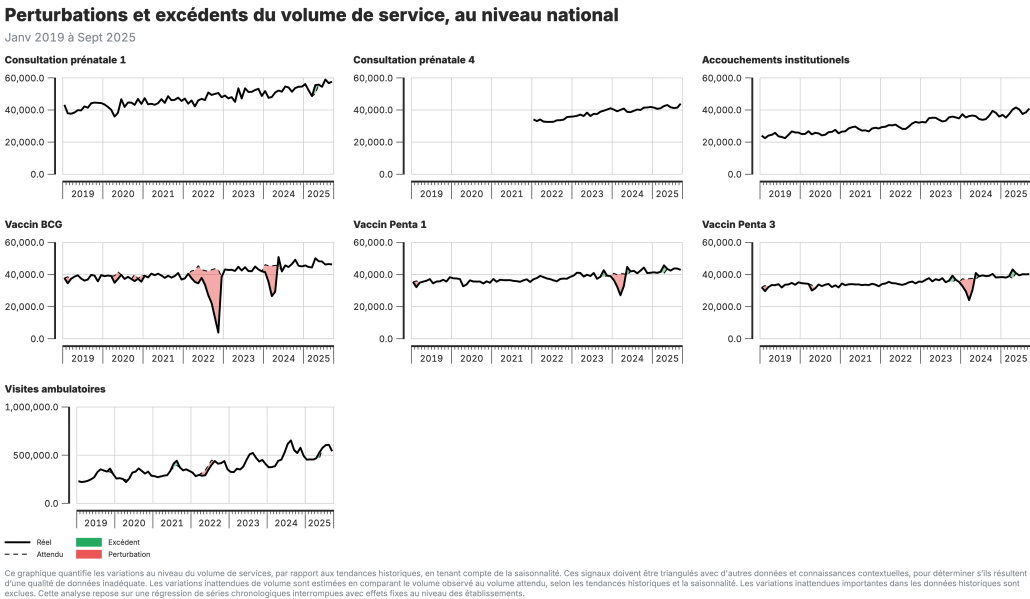


Chart interpretation

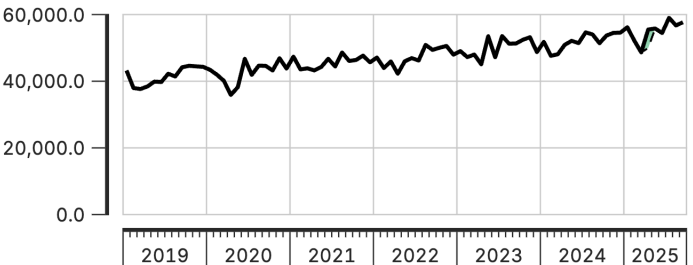
Element	Description
Black line	Actual (observed) service volumes
Red shaded areas	Shortfall periods (actual below expected)
Green shaded areas	Surplus periods (actual above expected)

The magnitude of the shaded area corresponds to the scale of the disruption.

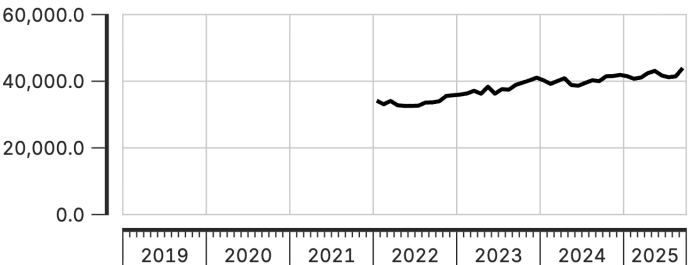
Perturbations et excédents du volume de service, au niveau national

Janv 2019 à Sept 2025

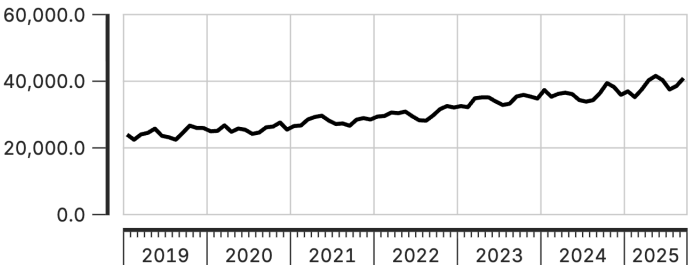
Consultation prénatale 1



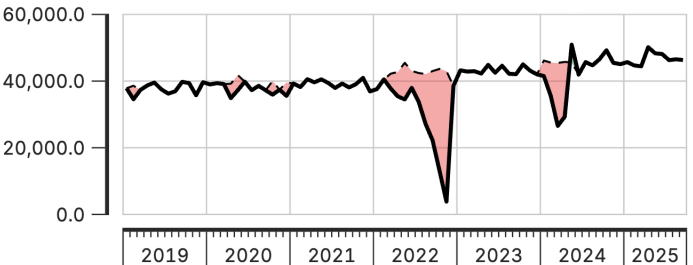
Consultation prénatale 4



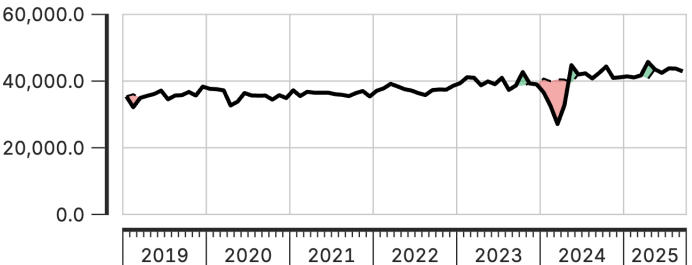
Accouchements institutionnels



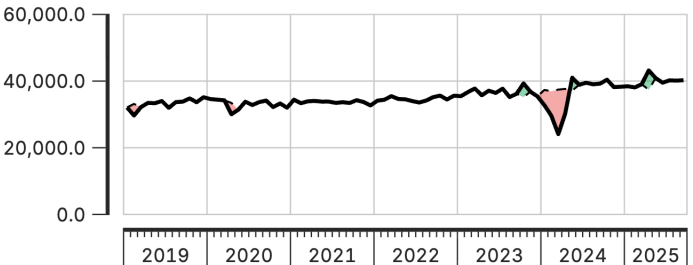
Vaccin BCG



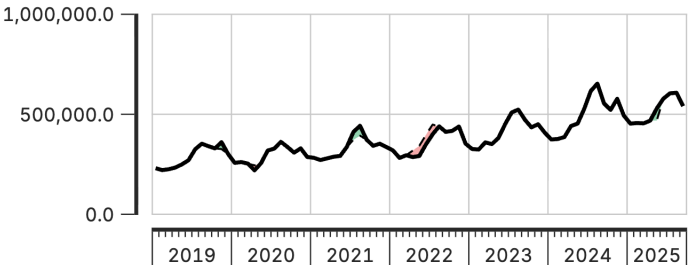
Vaccin Penta 1



Vaccin Penta 3



Visites ambulatoires

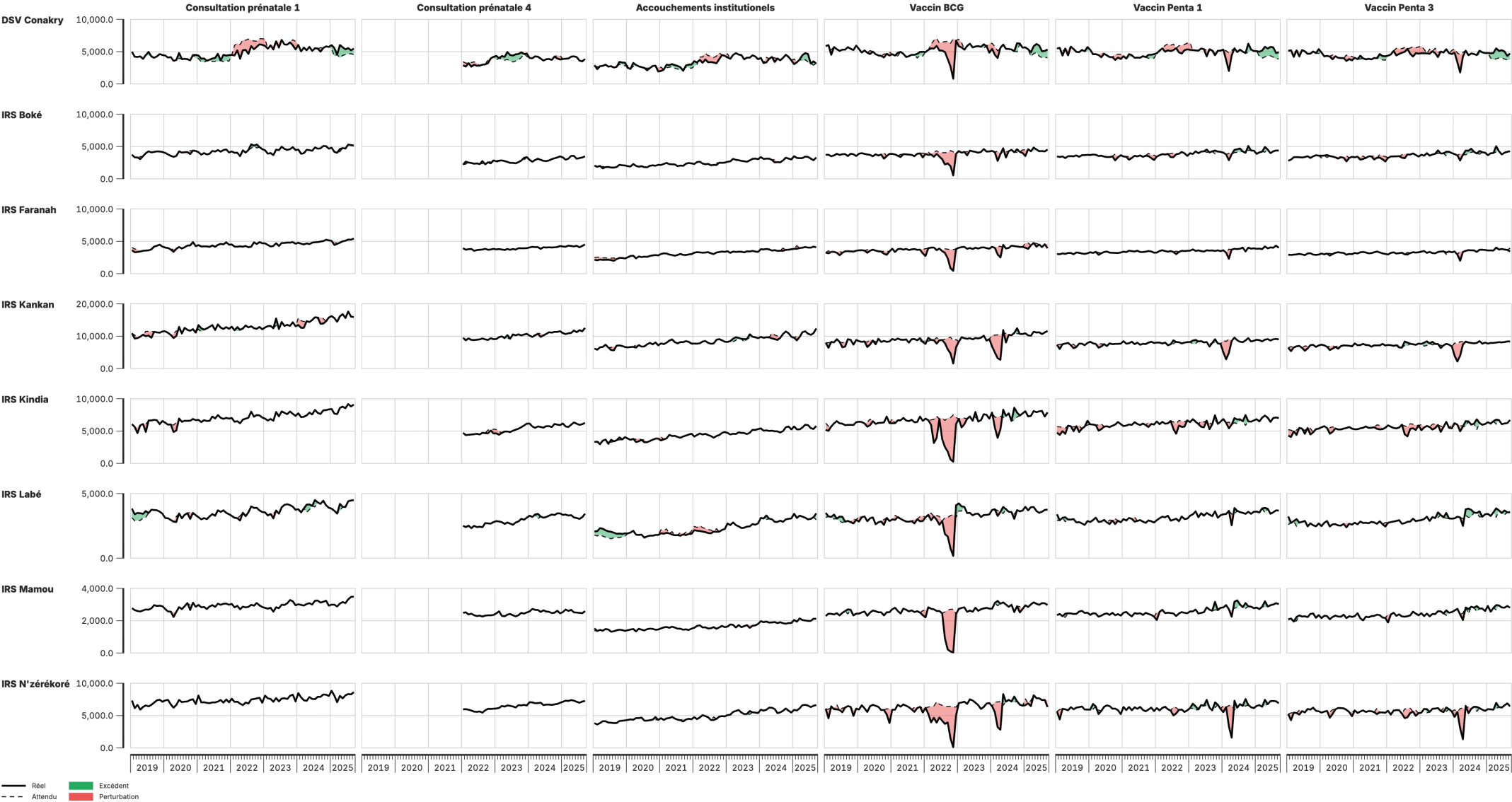


— Réel
 - - - Attendu
 ■ Excédent
 ■ Perturbation

Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

Perturbations et excédents du volume de service, au niveau infranational

Janv 2019 à Sept 2025



Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

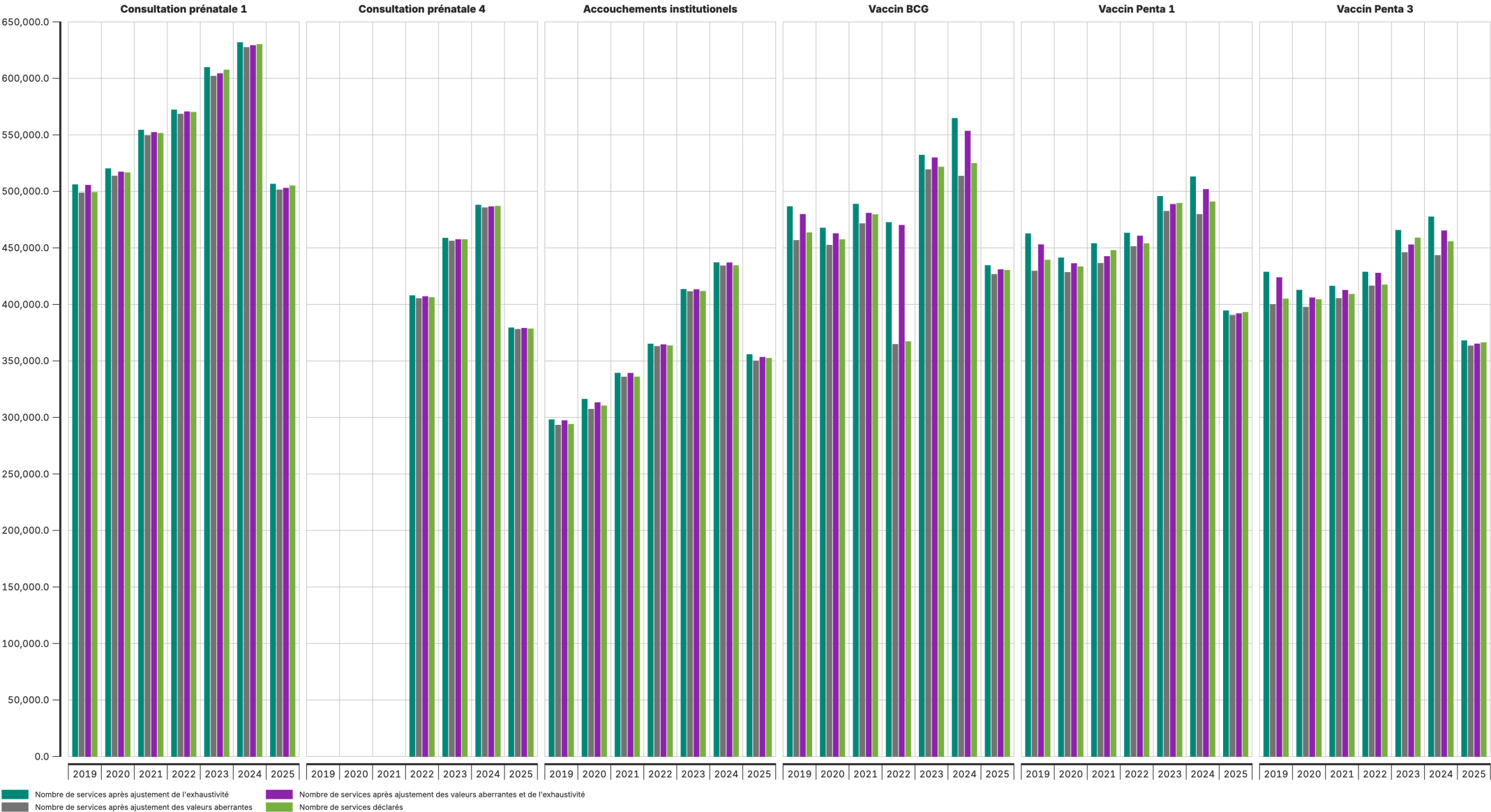
Data quality adjustment scenarios

The analysis supports four data versions to assess sensitivity to adjustment assumptions:

Scenario	Description
No adjustment	Raw reported values
Outlier adjustment	Extreme values corrected
Completeness adjustment	Adjusted for missing facility reports
Both adjustments	Outlier and completeness adjustments applied

Change in volume due to data quality adjustments

Janv 2019 à Sept 2025



Rationale for scenario comparison

Comparison across adjustment scenarios addresses three questions:

Question	Implication
Do adjustments substantially change results?	If results are similar across scenarios, findings are robust
Which adjustment has greater impact?	Identifies whether completeness or outlier correction drives differences
Are conclusions sensitive to assumptions?	Informs confidence in interpretation

Service utilization: Configuration parameters

Parameter	Default	Description
SELECTEDCOUNT	count_final_both	Data column used for analysis
SMOOTH_K	7	Rolling median window size (must be odd)
MADS_THRESHOLD	1.5	MAD threshold for sharp disruption detection
DIP_THRESHOLD	0.90	Flag periods below 90% of expected volume
DIFFPERCENT	10	Percentage threshold for visualization
RUN_DISTRICT_MODEL	FALSE	Enable district-level regression analysis
RUN_ADMIN_AREA_4_ANALYSIS	FALSE	Enable ward-level analysis

Tuning sensitivity

Objective	Adjustment
More sensitive detection	Lower <code>MADS_THRESHOLD</code> to 1.0; set <code>DIP_THRESHOLD</code> to 0.95; reduce <code>SMOOTH_K</code> to 5
Less sensitive detection	Increase <code>MADS_THRESHOLD</code> to 2.0; set <code>DIP_THRESHOLD</code> to 0.80; increase <code>SMOOTH_K</code> to 9 or 11
Faster runtime	Set <code>RUN_DISTRICT_MODEL</code> and <code>RUN_ADMIN_AREA_4_ANALYSIS</code> to FALSE
Detailed subnational analysis	Set <code>RUN_DISTRICT_MODEL</code> to TRUE (slower)

Service coverage estimates

The Coverage Estimates module estimates health service coverage: the percentage of the target population that received a given health service.

Data sources:

Source	Purpose
HMIS	Adjusted health service volumes
UN WPP	Population projections
DHS/MICS	Household survey benchmarks

| Two-part analytical process

Part	Components
Part 1: Denominator calculation	Calculate target populations using multiple methods; compare against survey benchmarks; select optimal denominator for each indicator
Part 2: Coverage estimation	Apply denominator selections; project survey estimates forward using HMIS trends; generate final coverage estimates

Definition of service coverage

Service coverage represents the proportion of the target population that received a specified health service.

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

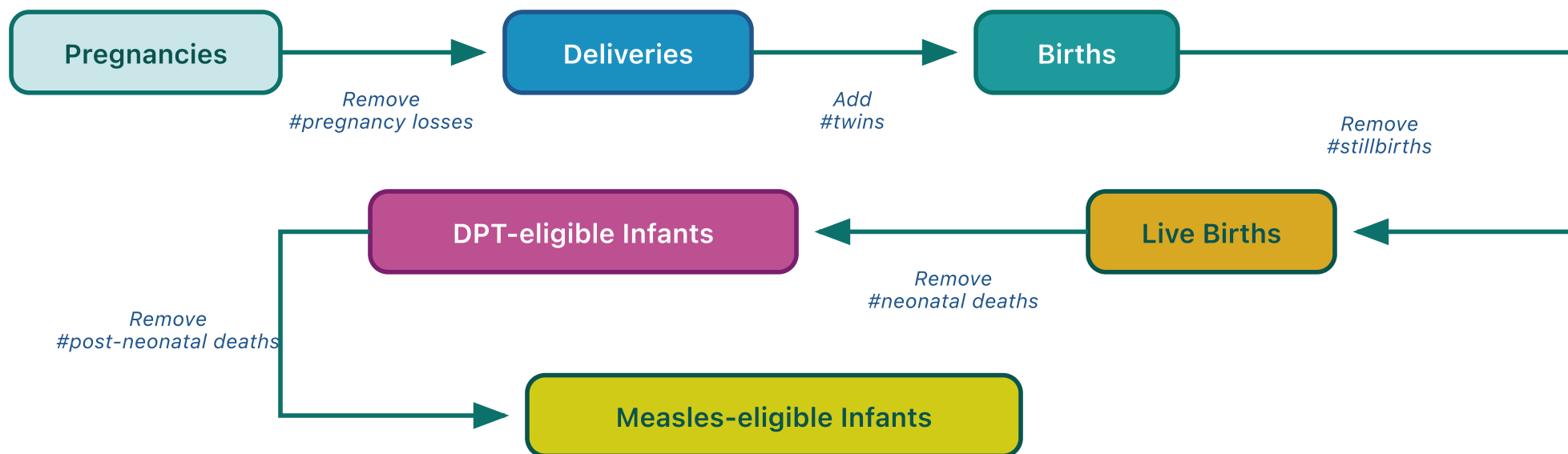
Denominators by service type

Each health indicator corresponds to a specific target population:

Service	Target population (denominator)
ANC1, ANC4	Pregnant women
Institutional delivery	Live births
BCG	Live births
Penta1, Penta3	Infants surviving beyond neonatal period
Measles1, Measles2	Infants surviving beyond infancy

Demographic cascade

Sequential demographic adjustments transform one target population estimate into another. Starting from pregnancies, demographic factors are applied to derive subsequent denominators:



Default values:

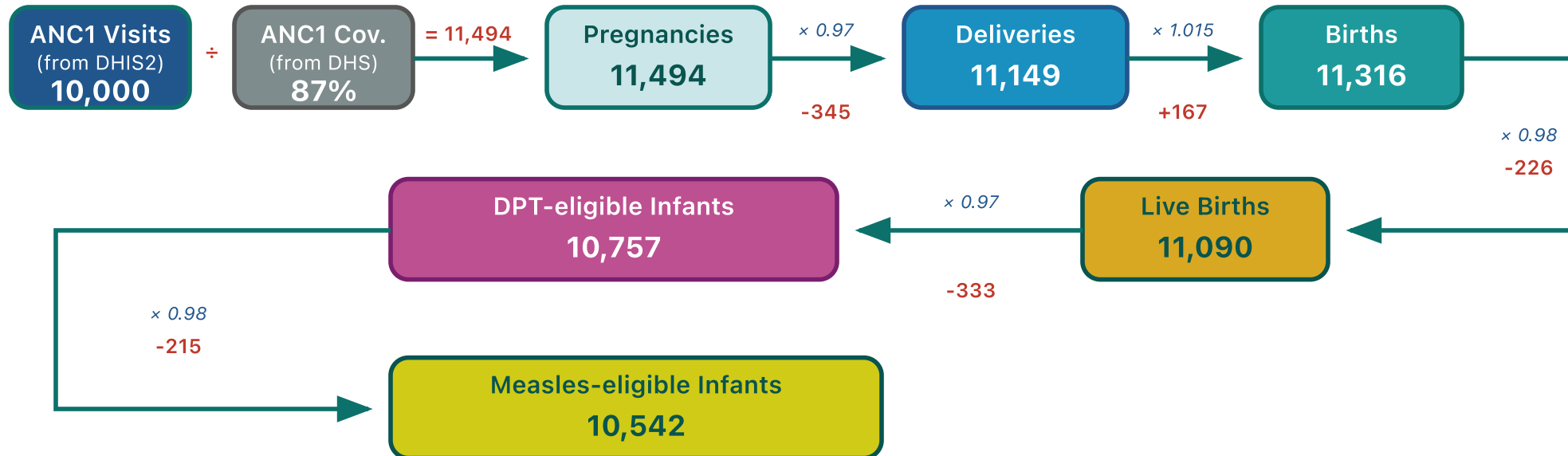
Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Denominator cascade: Illustration

Starting from ANC1 service counts, demographic adjustment factors are applied sequentially to derive denominators for other services:

Example: Calculating denominators from ANC1 visits



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = $10,000 \div 0.87 = 11,494$

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) → Deliveries (11,149) → Births (11,316) → Live Births (11,090) → DPT-eligible (10,757) → Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Forward and backward derivation

From any entry point, the cascade derives denominators in both directions:

Direction	Method	Example from Penta1
Forward	Apply mortality/attrition rates	DPT-eligible → Measles1-eligible → Measles2-eligible
Backward	Reverse mortality rates (add deaths back)	DPT-eligible → Live births → Births → Deliveries → Pregnancies

Backward derivation enables estimation of upstream populations from downstream service counts.

Denominators by entry point

Each HMIS indicator serves as an entry point. The module derives all target populations via forward and backward cascades:

Entry point	Base calculation	Forward derivation	Backward derivation
ANC1	ANC1 ÷ coverage → Pregnancies	Deliveries → Live births → DPT-eligible → Measles-eligible	—
Deliveries	Deliveries ÷ coverage → Deliveries	Live births → DPT-eligible → Measles-eligible	Pregnancies
BCG	BCG ÷ coverage → Live births	DPT-eligible → Measles-eligible	Deliveries → Pregnancies
Penta1	Penta1 ÷ coverage → DPT-eligible	Measles1-eligible → Measles2-eligible	Live births → Births → Deliveries → Pregnancies
UN WPP	Population projections	All denominators directly	—

Automatic denominator selection

For each indicator, the module selects the denominator that produces coverage closest to the survey benchmark.

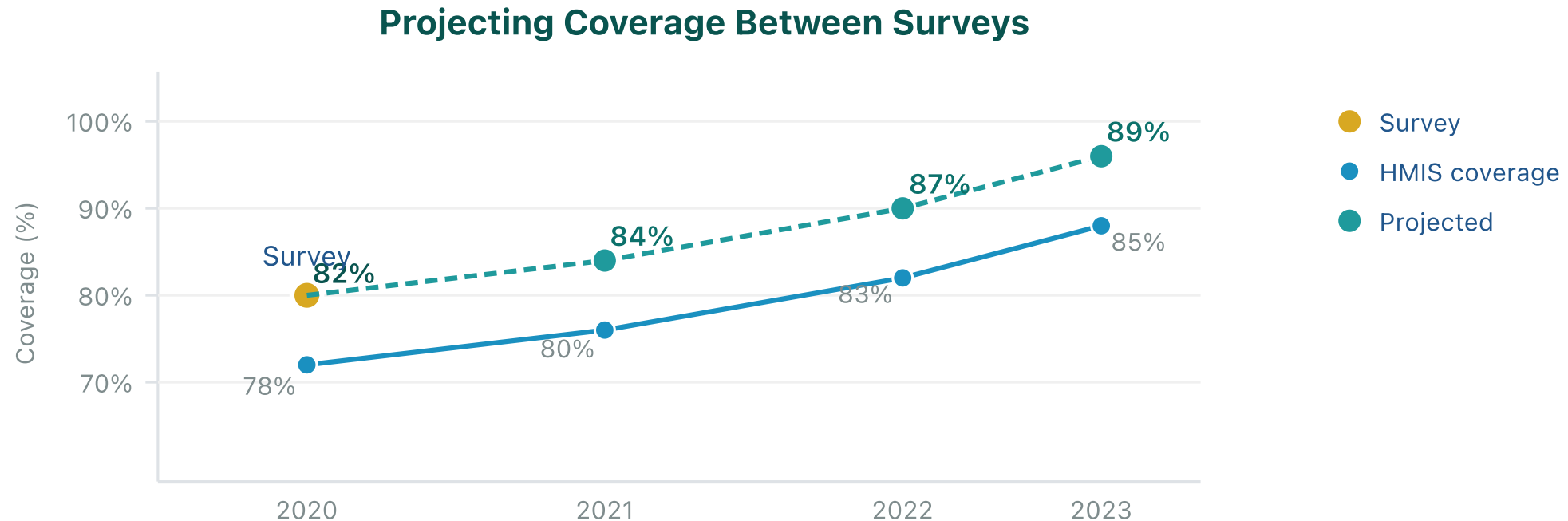
Selection algorithm:

1. Calculate coverage using each denominator option
2. Calculate squared error against survey: $(coverage - survey)^2$
3. Apply selection hierarchy (HMIS-based denominators prioritized over UN WPP)
4. Select the HMIS-based denominator with minimum error

Selection is made per indicator and geographic area. Users may override automatic selections in Part 2.

Coverage projection methodology

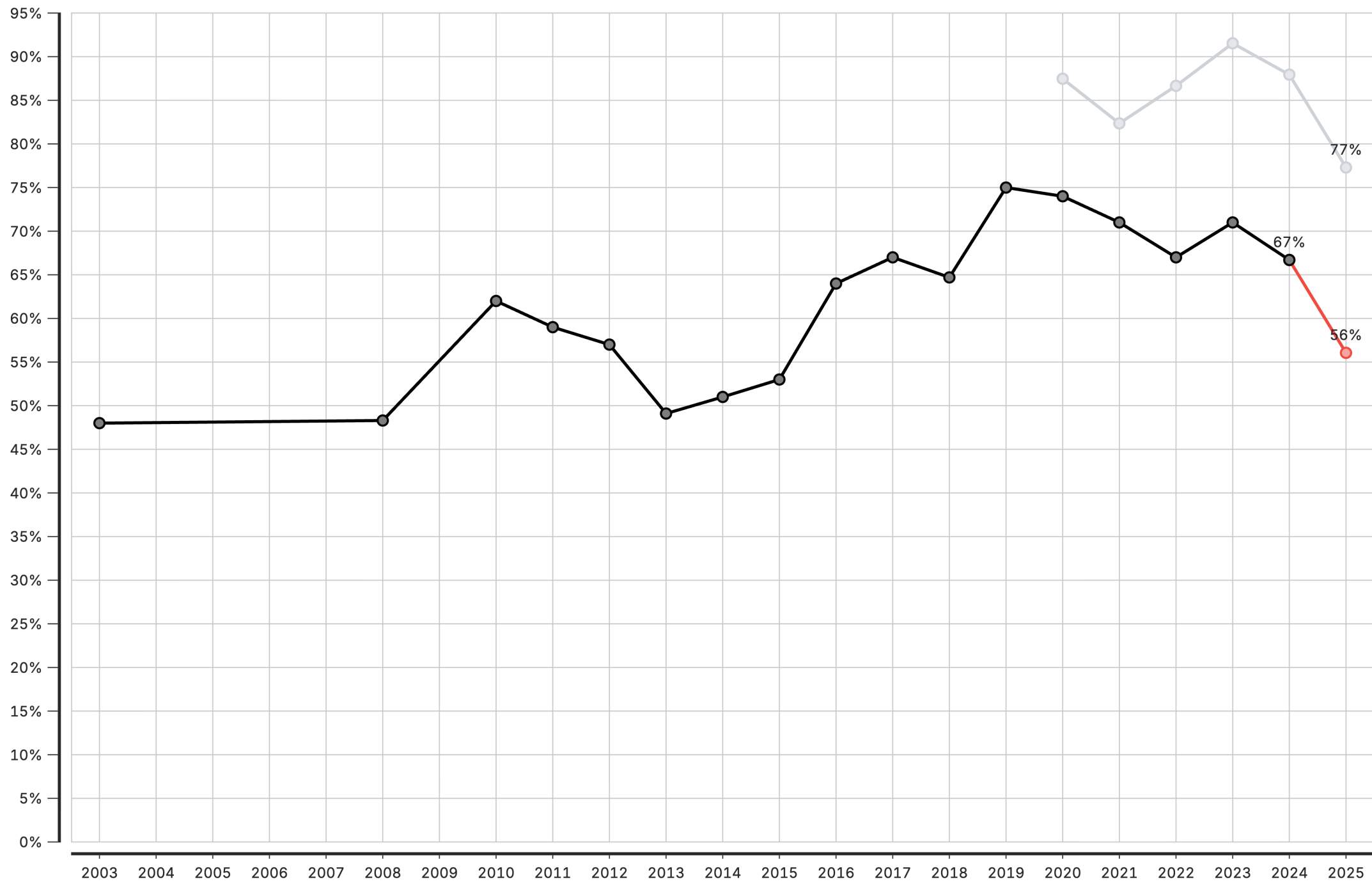
The module projects the most recent survey value forward using trends observed in HMIS-derived coverage:



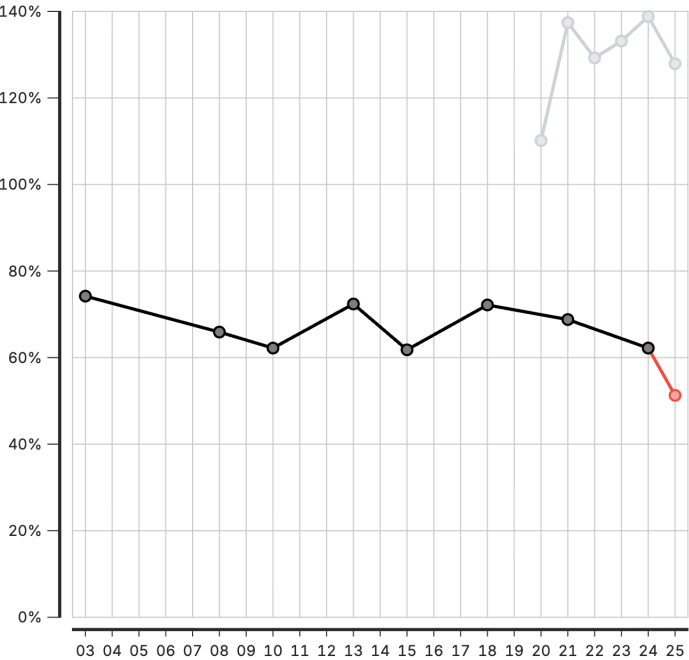
Year-over-year changes (deltas) in HMIS coverage are calculated and applied to the last survey value. This approach preserves the survey baseline while incorporating observed service delivery trends.

Interpretation of coverage outputs

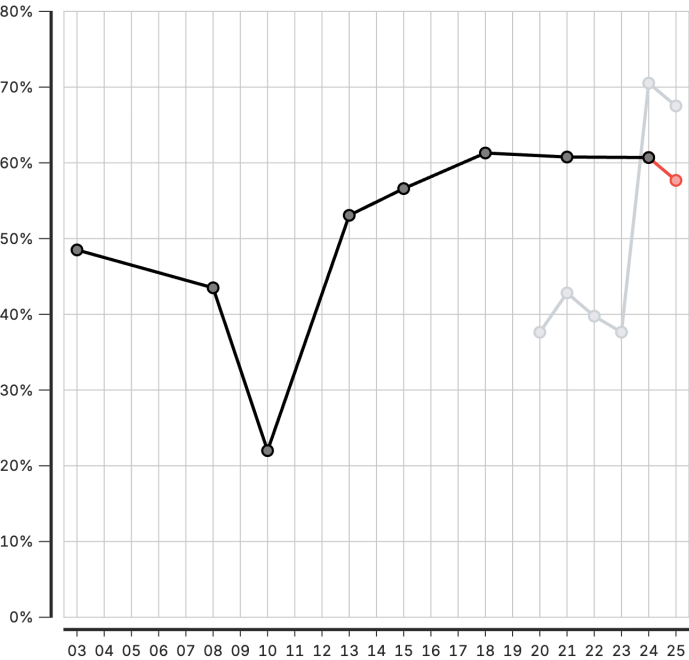
Element	Description
Black line/points	Survey data (DHS/MICS) — household survey reference
Grey line/points	HMIS-based coverage from facility data
Red line/points	Projected coverage — survey estimates extended using HMIS trends



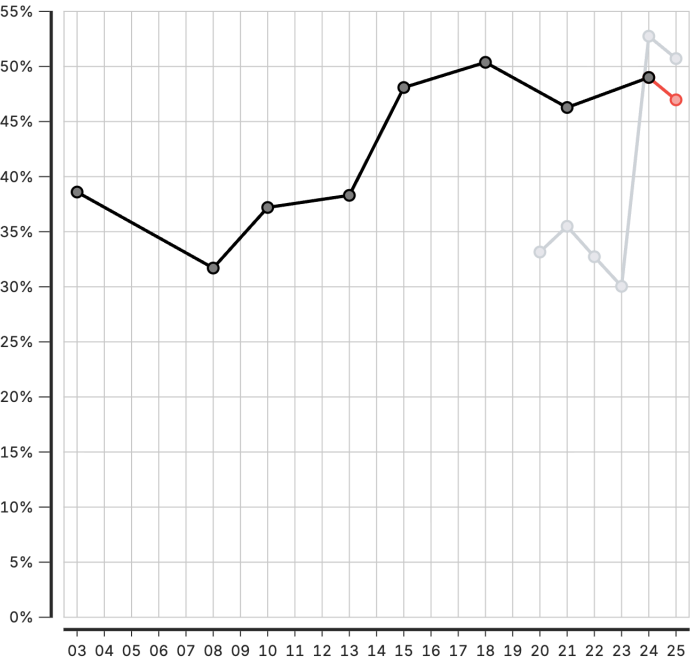
North Central Zone



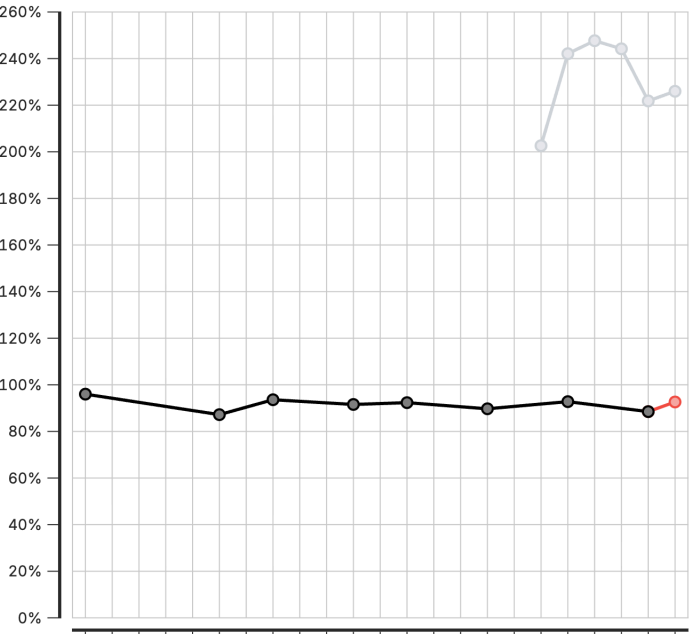
North East Zone



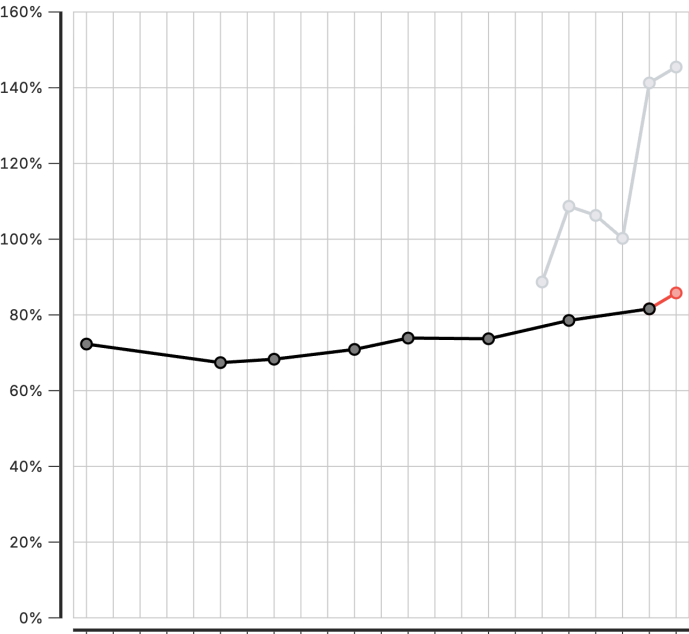
North West Zone



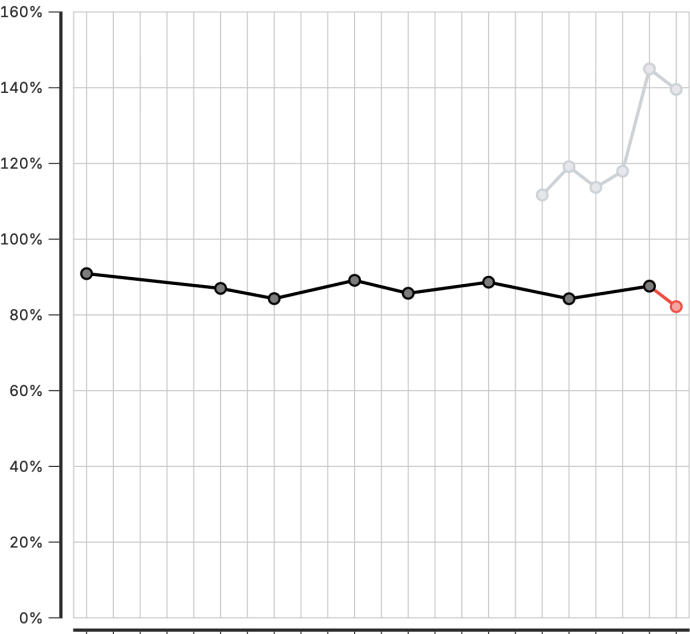
South East Zone



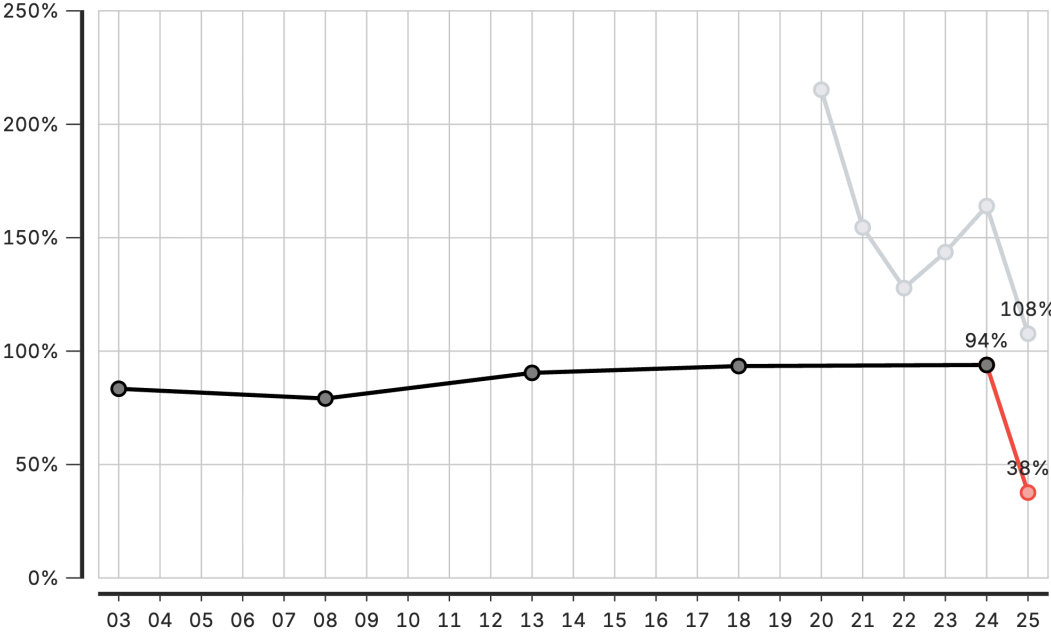
South South Zone



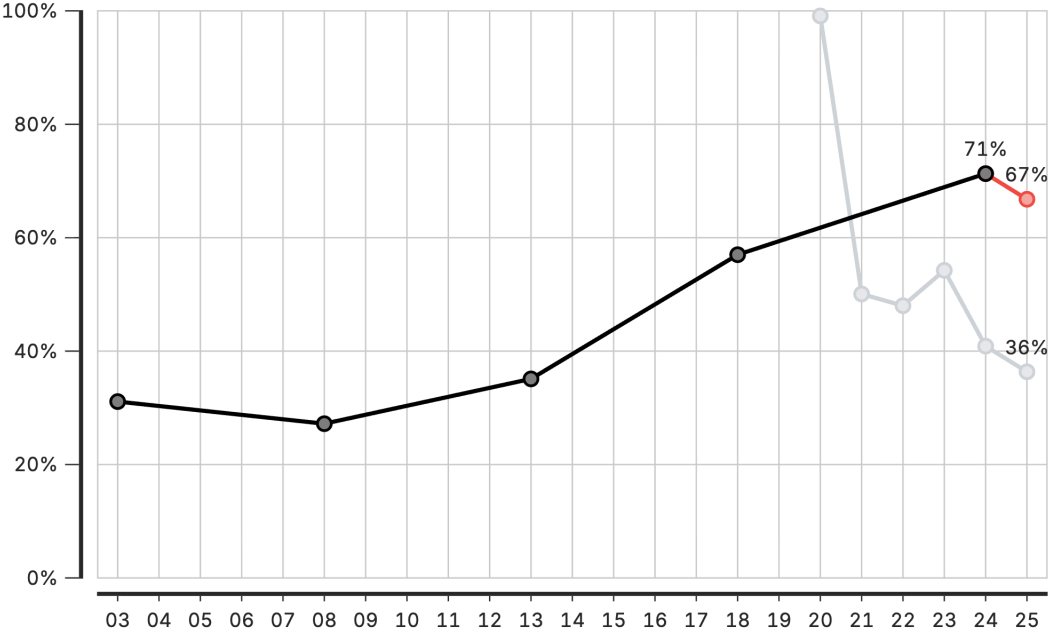
South West Zone



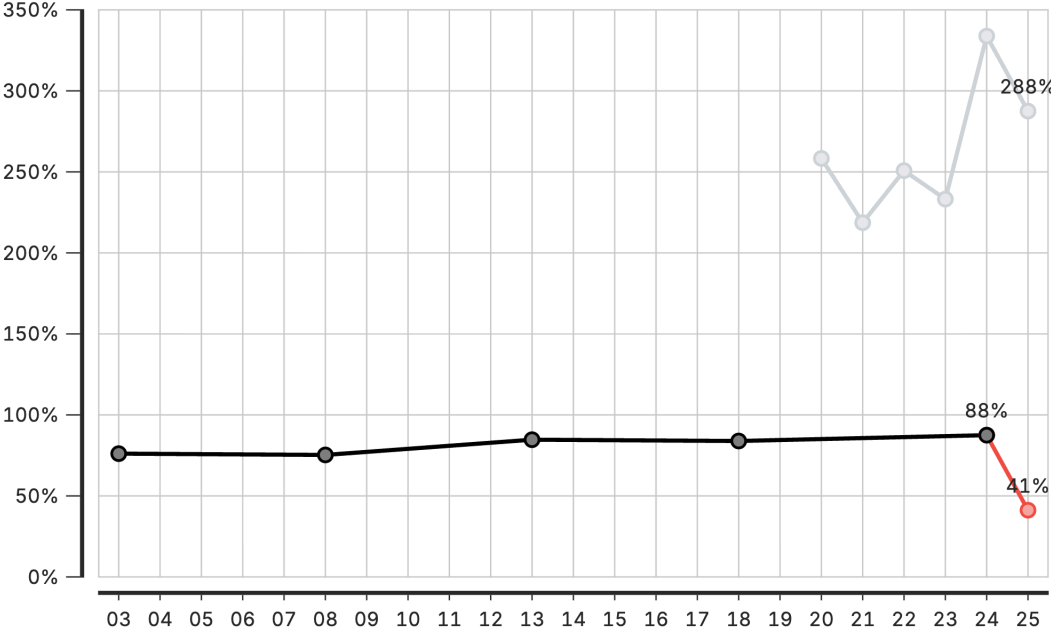
ab Abia State



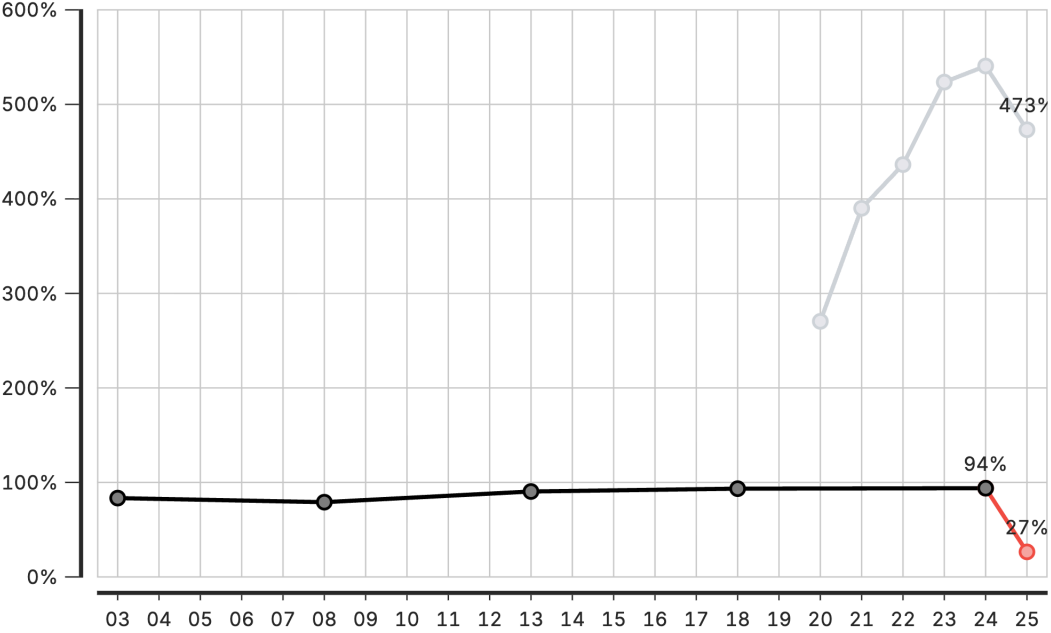
ad Adamawa State



ak Akwa-Ibom State



an Anambra state



Coverage module: Configuration parameters

Parameter	Default	Description
ANALYSIS_LEVEL	NATIONAL_PLUS_AA2	Geographic scope for analysis
SELECTED_COUNT_VARIABLE	count_final_both	Which adjusted count to use

Analysis level options:

- NATIONAL_ONLY — National analysis only
- NATIONAL_PLUS_AA2 — National + provinces/regions
- NATIONAL_PLUS_AA2_AA3 — National + provinces + districts

Demographic adjustment parameters

These parameters adjust denominators for the target population:

Parameter	Default	Description
PREGNANCY_LOSS_RATE	0.03	3% pregnancy loss
TWIN_RATE	0.015	1.5% twin births
STILLBIRTH_RATE	0.02	2% stillbirths
P1_NMR	0.039	Neonatal mortality rate
P2_PNMR	0.028	Post-neonatal mortality rate
INFANT_MORTALITY_RATE	0.063	Infant mortality rate
UNDER5_MORTALITY_RATE	0.103	Under-5 mortality rate

Note: Country-specific rates may be obtained from DHS reports, UN IGME, or national vital statistics.

Session 6: Creating a project

Activity: Creating a Project

In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Session 7: Creating visualizations

Activity: Creating Visualizations

In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis



Lunch Break

90 minutes

Back at 14:00

Session 8: Creating reports

Activity: Creating Reports

In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 3

Recap

On Day 2, we covered:

- FASTR methods for data quality assessment and adjustment
- Creating and configuring projects in the platform
- Building visualizations from Zambia data
- Generating reports

Day 3 Focus

Today we move from analysis to action:

- Deep dive into interpretation of results
- Create the Q4 2025 quarterly report
- Present findings to the group
- Develop action plans for continued work

Session 8: Interpretation of visualizations

Analytical thinking & interpretation

Content to be developed

This section will cover:

- Frameworks for interpreting FASTR outputs
- Connecting data patterns to programmatic meaning
- Common interpretation pitfalls to avoid
- Building analytical thinking skills

Session 9: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms



Lunch Break

90 minutes

Back at 14:00

Session 10: Presenting reports

End user mapping

End user mapping helps ensure that our outputs will meet the real needs of our end users.

| Key questions

1. Who is my end user?
2. What does this end user need to accomplish with the report?
3. What information are they most interested in?
4. What do they like/not like about current reports?
5. How do they like to receive their information?

Day 4

Recap

On Day 3, we covered:

- Interpretation of FASTR visualizations
- Creation of the Q4 2025 quarterly report
- Presentation of findings and group feedback
- Action planning for continued platform use

Day 4 Focus

Today we design the Health Facility Assessment:

- Overview of the FASTR HFA phone survey
- Review questionnaire structure
- Adapt questionnaire for Zambia context
- Plan HFA priorities and data use

Session 12: Overview of FASTR HFA phone survey

Health Facility Assessment Phone Survey

Placeholder: Content to be developed



Lunch Break

90 minutes

Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session - Zambia
January 27-30, 2026
Lusaka

Contact Information

FASTR Team

 Email:

 **Website:** <https://www.globalfinancingfacility.org/>

